

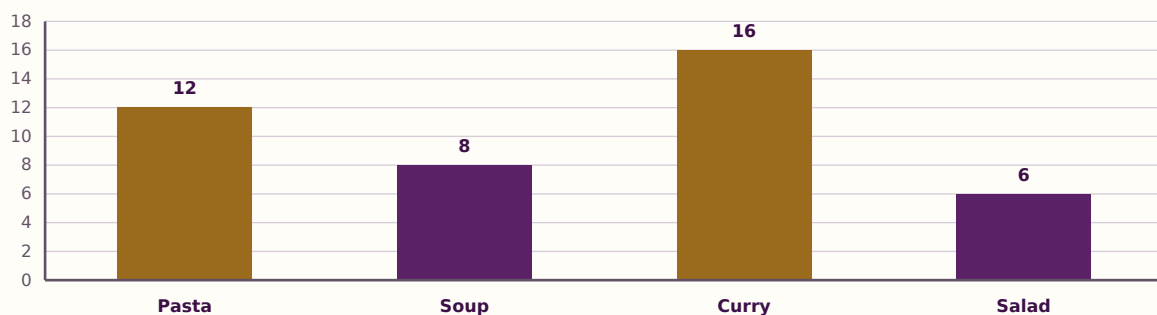
Reading and presenting data

Statistics is the process of collecting, organising, presenting and interpreting information. In Year 3, pupils use bar charts, pictograms and tables, then answer one-step and two-step questions about the data.

Bar charts make comparisons visible

This chart shows pupils' lunch choices. The vertical scale increases in steps of 2, so every gridline must be read carefully.

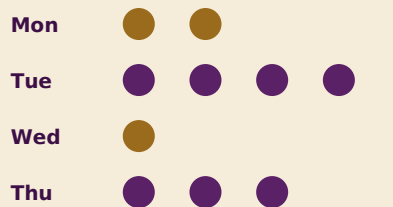
PUPILS CHOOSING EACH LUNCH



Pictograms use a key

A symbol may represent more than one item. Always read the key before counting or comparing.

RAINFALL PICTOGRAM



KEY: each circle represents 2 mm of rain

READ THE DISPLAY

Wettest day	Tuesday: 8 mm
Driest day	Wednesday: 2 mm
Difference	$8 \text{ mm} - 2 \text{ mm} = 6 \text{ mm}$

WHY SCALES MATTER

A bar reaching 8 on a scale of 2 represents 8 pupils - not four grid spaces.



Thinking with data

Reading a display is only the beginning. Pupils use the information to compare categories, combine values, find differences and explain what the data shows.

Tables organise exact values

WEEKLY READING TOTALS

Class	Books read
A	18
B	24
C	15
D	21

ONE-STEP COMPARISON

How many more did B read than C?

$$24 - 15 = 9$$

TWO-STEP QUESTION

A and D together: how many more than C?

$$18 + 21 - 15 = 24$$

Statistics in everyday life

TRAIN TABLE

choose a suitable departure time

LEAGUE TABLE

compare points or goals

WEATHER CHART

compare temperature or rainfall

Ask questions, not just read answers

Good statistical thinking asks what a pattern means: Which category is greatest? How many more? What changed? Is the scale fair? Does the available data support the conclusion?

NOTICE

What can I see?

COMPARE

What is greater or fewer?

CALCULATE

What totals or differences help?

CONCLUDE

What does the data show?

By the end of this topic, pupils should be able to:

- interpret and present bar charts
- read and construct simple tables
- answer one-step comparison questions
- interpret and present pictograms
- read scales of 2, 5 and 10 accurately
- answer two-step questions using totals and differences

LOOKING AHEAD: Statistics supports science, geography, computing and informed decisions.